MIL-C-14385B(WC)
19 July 1972
SUPERSEDING
MIL-C-14385A(WC)
20 March 1967

MILITARY SPECIFICATION

CANNON, 40MM DUAL AUTOMATIC GUN: M2A1
AND CLIP, CHARGER: M3

1. SCOPE

1.1 This specification covers a recoil-operated, dual cannon for single and automatic firing and a charger-clip for feeding the cartridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal
TT-E-489 - Enamel; Alkyd, Gloss for Exterior and Interior Surfaces

Military
MIL-C-13931 - Cannons, General Specifications for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Federal
FED-STD-595 - Colors

Military
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspections by Attributes

PUBLICATIONS

U. S. Army Weapons Command
Gage, 5276650 - Gage, Firing Pin Protrusion
Gage, 7247039 - Gage, Headspace
Gage, 7247040 - Gage, Headspace
EPL 7305371 - Engineering Parts List for Cannon, 40MM Dual Automatic Gun: M2A1

FSC 1010

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Master LSQAP-7305371 - Master Index of Supplementary Quality Assurance Provisions for Cannon, 40MM Dual Automatic Gun: M2A1
DL 7305371 - Index of Inspection Equipment Data Lists for Cannon, 40MM Dual Automatic Gun: M2A1
PDS P 7305371 - Packaging Data Sheet, Gun, Automatic, 40MM, M2A1

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1), the first article shall be included in the pilot pack (see 5.1).

3.2 Materials and construction. The cannons and charger-clips shall conform to this specification, MIL-C-13931, the drawings and specifications listed in the Engineering Parts List EPL 7305371.

3.3 Operation.

3.3.1 Cannons. The cannons shall operate within the designed rate of fire without interference, erratic movement or malfunction.

3.3.2 Charger-clips. The assembled clips shall hold each round firmly in proper position. Each hook of the assembled clips shall move to extended position under applied pressure and return to retraction position under spring pressure without lag, sticking or binding.

3.4 Mechanical requirements.

3.4.1 Breech casing bodies. Preparatory to machining the fitted bolt holes, the breech casings shall be so assembled in matched pairs as to insure correct alignment of the trunnions and the recoil guides.

3.4.2 Firing pins. The point of the firing pins may be finished with crocus cloth but shall not be hand stoned. The protrusion of the firing pin from the face of the breechblock shall be not less than 0.099 inch or greater than 0.114 inch.

3.4.3 Headspace. Headspace shall be not less than 0.002 inch or greater than 0.024 inch. Preferable clearance shall be between 0.004 inches and 0.014 inches. (Headspace is the clearance between the forward face of the breechblock and the base of a maximum cartridge when both are in the firing position.)



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3.5 Lubrication fittings. The lubrication fittings shall be painted with red, gloss, alkyd enamel conforming to Class A, Composition G, of TT-E-489. The color shall match No. 11136 of FED-STD-595.

3.6 Recoil cylinders. The recoil mechanism shall be filled with the type and volume of fluid specified on the recoil cylinder assembly drawing.

3.7 Proof testing. Cannons, repair tubes, and flash suppressors shall be subject to proof testing.

3.7.1 Performance. Cannons shall function properly during single and automatic firing. Irregularities and defects shall be corrected before acceptance.

3.7.2 High pressure resistance. The cannon shall be capable of withstanding a pressure of 113 ± 4 percent of the 43,000 pounds per square inch upper pressure limit when tested as specified in 4.5.2.3.

3.7.3 Firing rate. The firing rate on AUTO-FIRE shall be not less than 120 rounds per minute from each of the cannons, at approximately 70°F ambient temperature.

3.7.4 Recoil. The recoil cylinder of each cannon shall be adjusted so that the recoil shall be not less than 7.4 inches nor more than 8.3 inches.

3.7.5 Material soundness. After proof firing, all parts shall be free of cracks and fractures and shall comply with the soundness requirements specified on the applicable documents.

3.7.6 Bore enlargement. Bore diameters, measured across the rifled lands after proof firing no more than 15 rounds, shall not exceed the inside (minimum) diameter specified on the applicable drawing by more than the enlargement specified in Table I.

Table I Bore Enlargement

| Distance forward of commencement of rifling | 0.10 inch | 1 cal. | 2 Cal. to muzzle |
|---|---|---|---|
| Enlargement (inch) | 0.004 | 0.004 | 0.003 |

NOTE: Interpolate linearly to determine the allowable bore enlargement at distances other than the distances given.

3.8 Workmanship. Workmanship requirements shall be in accordance with MIL-C-13931.

4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article (see 4.4).
b. Quality conformance (see 4.5).

4.4 First article inspection. A first article shall be submitted for inspection (in accordance with the contract requirements). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Cannons and charger-clips shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAPs listed in Master LSQAP 7305371 and SQAP-APPENDIX-WVT.

4.5.1.2 Breech casing bodies. Alignment of the trunnions and the recoil guides shall be gaged to assure compliance with 3.4.1.

4.5.1.3 Firing pin protrusion. The protrusion of the firing pin from the face of the breechblock shall be measured with firing pin protrusion gage 5276650 or an approved equivalent, to determine compliance with the requirement of 3.4.2.

4.5.1.4 Headspace. Headspace shall be measured with Headspace gages 7247039 and 7247040 or an approved equivalent to determine compliance with the requirement of 3.4.3.



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4.5.1.5 Lubrication fittings. Lubrication fittings shall be examined to determine conformance with the painting requirement of 3.5.

4.5.1.6 Recoil cylinders. The recoil mechanism shall be examined to assure it is filled with the proper type and volume of fluid as specified in 3.6.

4.5.1.7 Workmanship. Workmanship shall be examined in accordance with MIL-C-13931.

4.5.1.8 Final examination. Final examination of each cannon and each charger-clip shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.1.9 Inspection of preparation for delivery.

4.5.1.9.1 Examination. The classification of defects and the AQLs for examination of the packaging, packing, and marking are as follows:

| Category | Defect | AQL | Insp. Method |
|---|---|---|---|
| Major | | | |
| 101 | Illegible or incorrect marking | 1.0 | Visual |
| 102 | Improper level of preservation, packaging or packing | 1.0 | Visual |
| 103 | Inadequate cleaning or drying | 1.5 | Visual |
| 104 | Improper preservative application | 1.5 | Visual |
| 105 | Missing or improper cushioning | 1.5 | Visual |
| 106 | Improper blocking or bracing | 1.5 | Visual/Manual |
| 107 | Improper closure or strapping of container | 1.5 | Visual/Manual |
| Minor | | | |
| 201 | Workmanship | 4.0 | Visual |

4.5.1.10 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Tests.

4.5.2.1 Functioning of cannons. Each cannon shall be tested to assure smoothness of operation, and absence of interference, binding, erratic movement and malfunction. Failure to comply with the requirements of 3.3.1 shall be cause for rejection.

4.5.2.2 Functioning of charger-clips. Charger-clips shall be tested with dummy rounds to assure compliance with the functional requirements of 3.3.2.



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4.5.2.3 Proof testing.

4.5.2.3.1 Cannon. Unless otherwise specified in the contract, each cannon with flash suppressor shall be proof fired by Government personnel at a proving ground designated by TECOM in accordance with Table II.

Table II - Cannon Firing Schedule

| No. of Rounds | Percent of Upper Pressure Limit | Elevation (Deg.) |
|---|---|---|
| 1 | 113 ± 4 | 0 |
| 1 | 113 ± 4 | 75 |
| 8 (Auto-Fire) | 100 | 75 |

4.5.2.3.2 Repair tubes. Unless otherwise specified in the contract, repair tubes shall be proof fired on a sampling basis in accordance with MIL-STD-1235 with the following criteria as part of the sampling plan:

a. Inspection Level: II
b. Plan: CSP-1
c. Lot Size: One month's production
d. AQL: 0.10%

The firing schedule shall be in accordance with Table III.

4.5.2.3.3 Repair flash suppressors. Unless otherwise specified in the contract repair flash suppressors shall be proof fired on a sampling basis in accordance with 4.5.2.3.2 except the AQL shall be 1.0%. The firing schedule shall be in accordance with Table III.

Table III - Repair Tube and Repair Flash Suppressor Firing Schedule

| No. of Rounds | Percent of Upper Pressure Limit |
|---|---|
| 1 | 113 ± 4 |

So far as practical proof firing of repair flash suppressors shall be combined with that of repair tubes. If a combined test is impracticable, the flash suppressors shall be proof fired on a proof facility gun.

4.5.2.3.4 Performance. Cannons shall be inspected to assure proper functioning during single and automatic firing to comply with 3.7.1.

4.5.2.3.5 Firing rate. The firing rate on AUTO-FIRE shall be determined to assure conformance with 3.7.3. The firing rate shall be determined at 75° elevation at approximately 70°F ambient temperature.



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4.5.2.3.6 Recoil. The recoil shall be measured to determine conformance with 3.7.4.

4.5.2.3.7 Material soundness. After proof firing, the flash suppressor, breech ring, breechblock, and the exterior surface of the tube shall be magnetic particle tested in accordance with applicable drawings and specifications. In addition, the bore shall be examined with a borescope.

4.5.2.3.8 Bore enlargement. The bore diameters across the lands of the cannon tubes shall be measured after proof firing no more than 15 rounds. Measurements shall be made with a star gage, an air gage, or other approved instrument to determine compliance with 3.7.6 and Table I. Measurements shall be recorded.

4.5.2.3.9 Records. Cannons, repair tubes, and repair flash suppressors that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground. Repair tubes and repair flash suppressors accepted on a proof sampling basis shall be furnished with applicable data on DA Form 2408-4 by the contractor and approved by the Government representative.

4.5.3 Inspection equipment. Data lists containing equipment used in the inspection of parts, sub-assemblies, and assemblies are listed in DL 7305371.

4.5.3.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in the List of Inspection Equipment DL 7305371, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack shall be forwarded in accordance with 3.1. Pilot packs for the cannon and clips shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P7305371.

5.2 Preservation, packaging, and marking. The cannon and clips shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P7305371.



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5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number or units) of the first article sample.

(3) The contract should designate at whose expense a retest may be performed.

(4) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(5) The contract should specify that the acceptable first article sample shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification shall be specified and identified as a separate item of the contract.

c. Serial numbers for the cannons, repair tubes and repair flash suppressors.

d. List of inspection equipment, responsibility for acquisition, maintenance and disposition thereof, if other than as specified (see 4.5.3.1).

e. Extent of supplier's responsibility for Government-furnished equipment.



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f. Selection of applicable levels of preservation, packaging, and packing (see 5.2).

g. Packaging instructions for repair parts (see 5.3).

h. Applicable acceptance test procedures and frequency of testing (see 4.5.1 and 4.5.2).

i. The contract shall specify under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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